



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Faculty of Engineering, Built Environment and Information Technology

Fakulteit Ingenieurswese, Bou-omgewing en
Inligtingtegnologie / Lefapha la Boetšenere,
Tikologo ya Kago le Theknološi ya Tshedimošo



Unit 12: Time Value of Money (Present Worth Analysis)

BSS 310 Engineering Management 2026

Lecturer: Prof. Michael Ayomoh

Quick Recap/Questions arising from Previous Lecture

The Lecture Addressed:

- Rudimentary knowledge about the MARR including its driving factors and other peripherals
- Discussions on the structure and source of capital
- Ability to differentiate between debt and equity financing
- Ability to calculate the weighted average cost of capital
- How to set the MARR for the semester project

Learning Outcomes for the Current Lecture

At the end of this lecture you should be able to:

- **Define** and introduce the rudiments of cash flow diagrams
- **Discuss, analyse, apply, and evaluate** the time value of money (with focus on discounting to the present, to obtain the Present Value (PV) of money
- **Establish** practical relationship between the time value of money, the MARR and a Project's viability

Presentation Outline

- Define and introduce the rudiments of cash flow
- Calculate time value of money
- Link the Calculations to the MARR, interest rates in general and a project's viability
- Apply the entire knowledge to the semester project

Directional Nomenclature for a Cost-Revenue Flow

(receipts)

+ **Cash comes in**

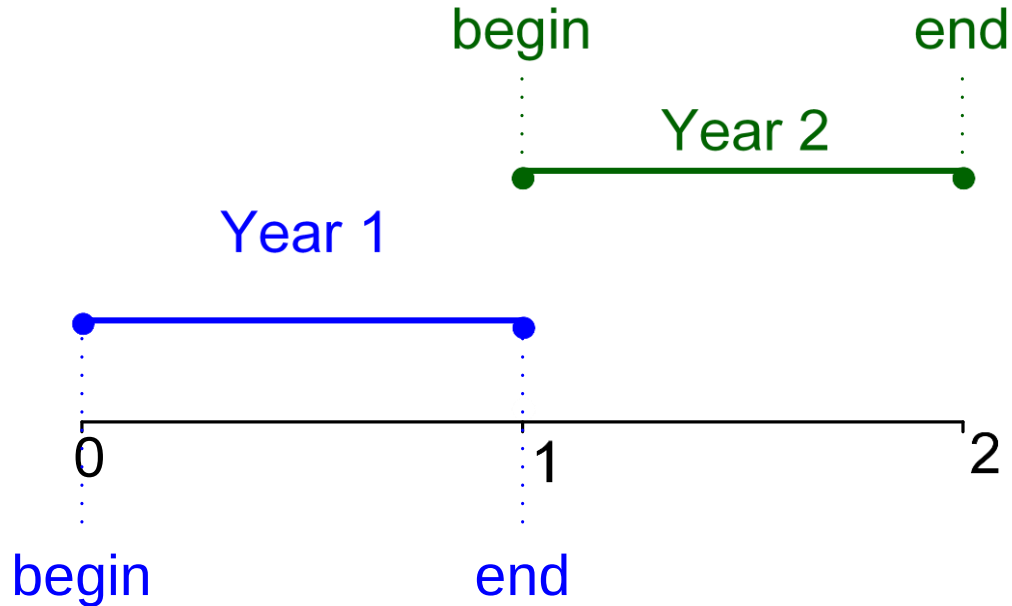
0 1 2 3 4 5

years

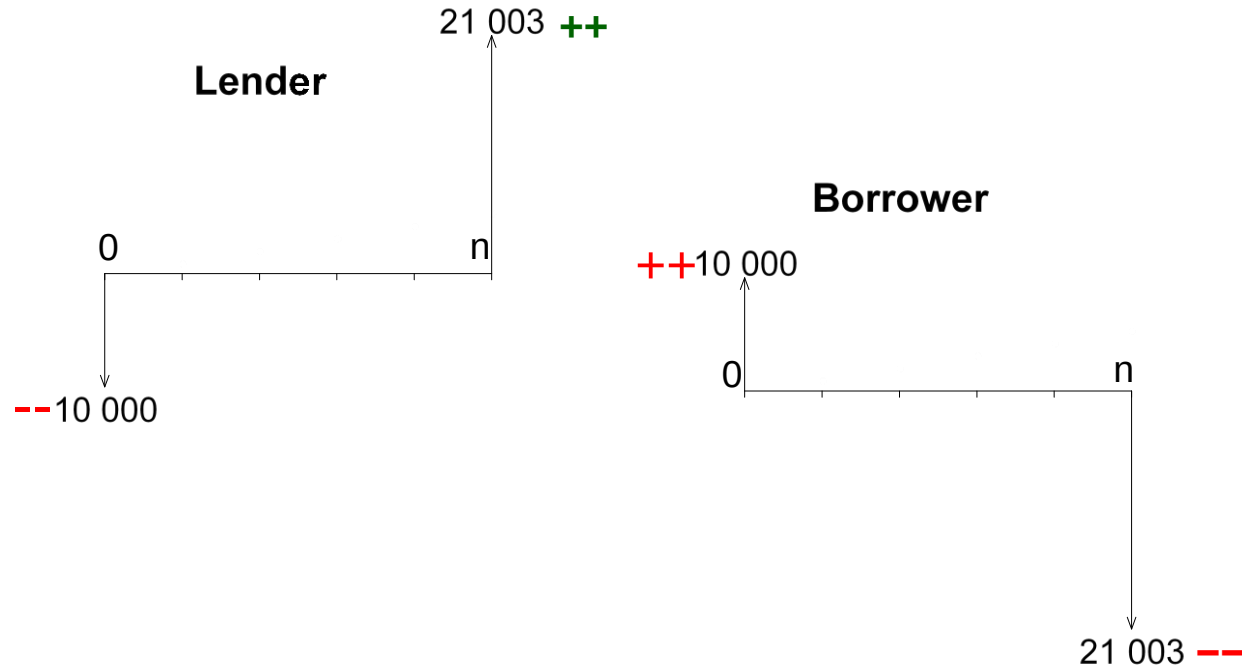
- **Cash goes out**

(disbursements)

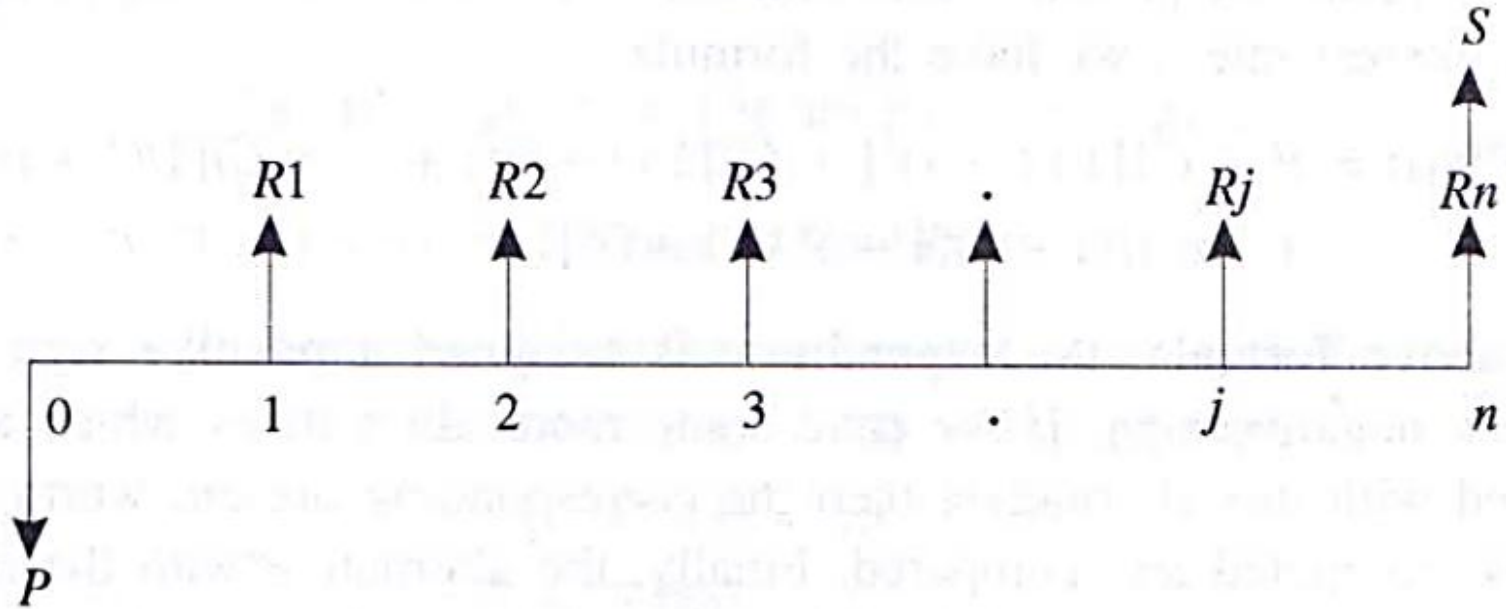
The Tail-Head Representation of a Transaction



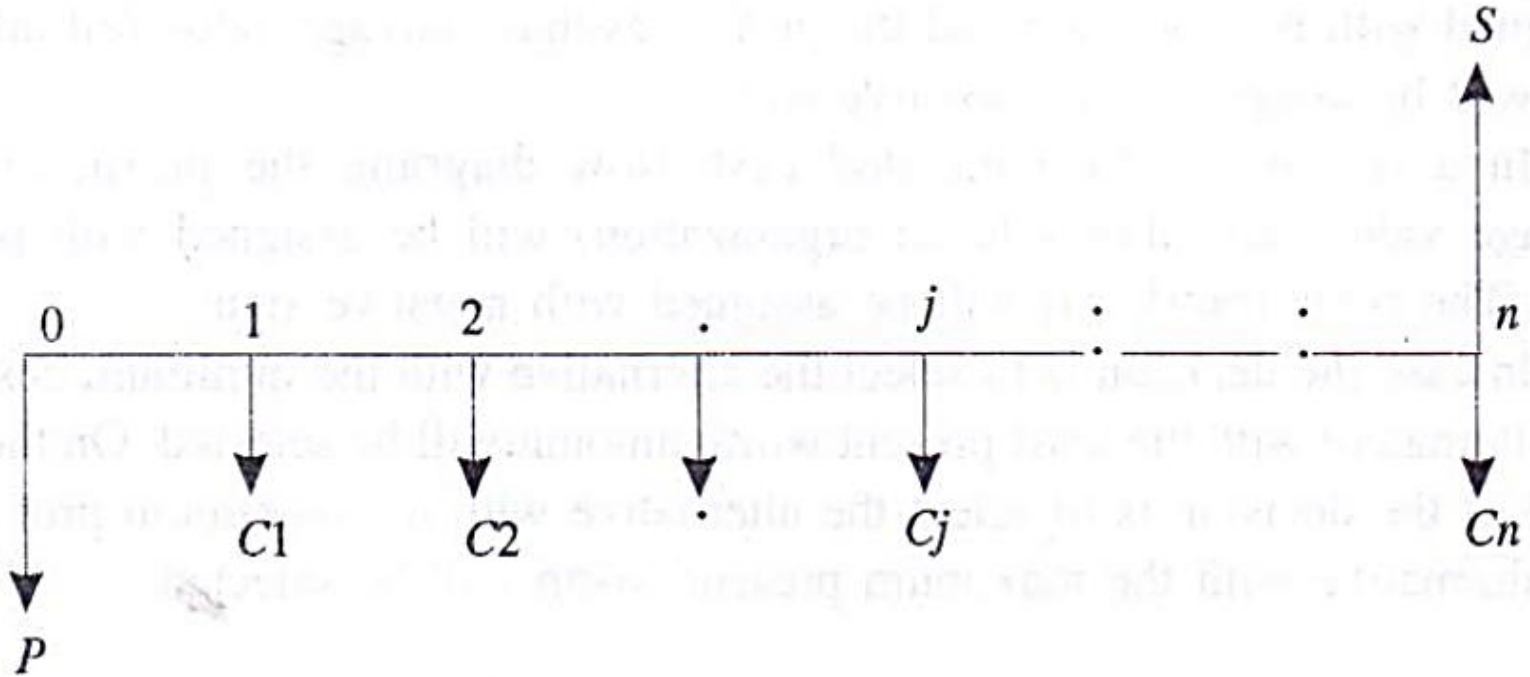
Debit and Credit Flow of a Cash Transaction



Revenue-Dominated Cash flow



Cost-Dominated Cash flow



Equations for Calculating the Time Value of Money

$$P = \frac{F}{(1+i)^n} = F(P/F, i, n)$$

$$F = P(1+i)^n = P(F/P, i, n)$$

$$A = F \frac{i}{(1+i)^n - 1} = F(A/F, i, n)$$

Equations for Calculating the Time Value of Money

$$F = A \frac{(1 + i)^n - 1}{i} = A(F/A, i, n)$$

$$A = P \frac{i(1 + i)^n}{(1 + i)^n - 1} = P(A/P, i, n)$$

$$P = A \frac{(1 + i)^n - 1}{i(1 + i)^n} = A(P/A, i, n)$$

Exercise #1: [Present Worth Analysis Determination]

If an investor invests a sum of R100 in a fixed deposit for 5 years with an interest rate of 15% compounded annually, the accumulated amount at the end of every year will be.....

End of year (n)	Interest accrued in (Rands)	Compound amount after n year(s)
0		100
1	15.00	115.00
2	17.25	132.25
3	19.84	152.09
4	22.81	174.90
5	26.24	201.14

Exercise #2:

Sequel to the previous problem, assuming we want R100 at the end of the n th year, what is the amount that should be deposited now, given an interest rate of 15%? Solve for the n th year = 2, 4, 5, 6, 10.

End of year (n)	Present worth	Compound amount after n year(s)
0		100
2	75.61	100
4	57.18	100
5	49.72	100
6	43.29	100
10	24.72	100

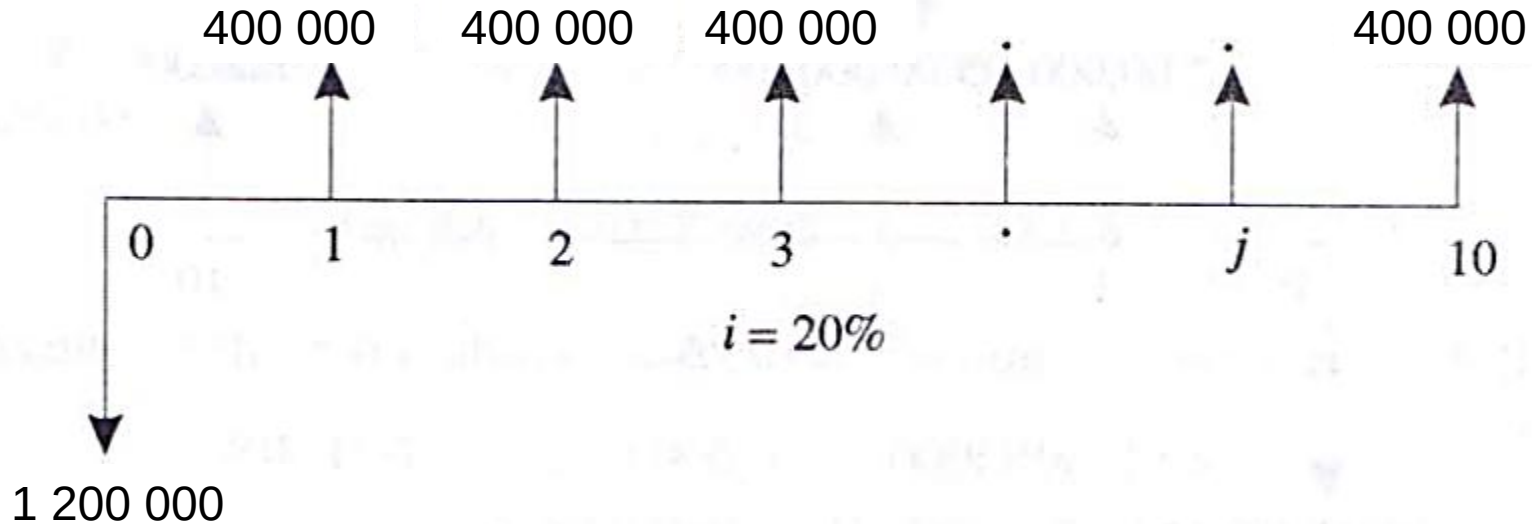
Example 1 [Using Cash Flow Diagrams: PV Discounting from the future]

VW South Africa is planning to expand its production operation. It has identified three different technologies for meeting the goal. The initial outlay and annual revenues with respect to each of the technologies are presented in the table below. Suggest the best technology which is to be implemented based on the present worth method of comparison assuming 20% interest rate, compounded annually.

Example 1 Contd.....

	<i>Initial outlay</i>	<i>Annual revenue</i>	<i>Life</i>
	R	R	(years)
Technology 1	1 200 000	400 000	10
Technology 2	2 000 000	600 000	10
Technology 3	1 800 000	500 000	10

Example 1: Cash Flow Diagram (Technology 1)



Example 1: Solution Contd...(Technology 1)

The present worth expression for this technology is

$$\begin{aligned}PW(20\%)_1 &= -1\,200\,000 + 400\,000 \times (P/A, 20\%, 10) \\&= -1\,200\,000 + 400\,000 \times (4.1925) \\&= -1\,200\,000 + 1\,677\,000 \\&= R477\,000\end{aligned}$$

Example 1: (Technology 2)

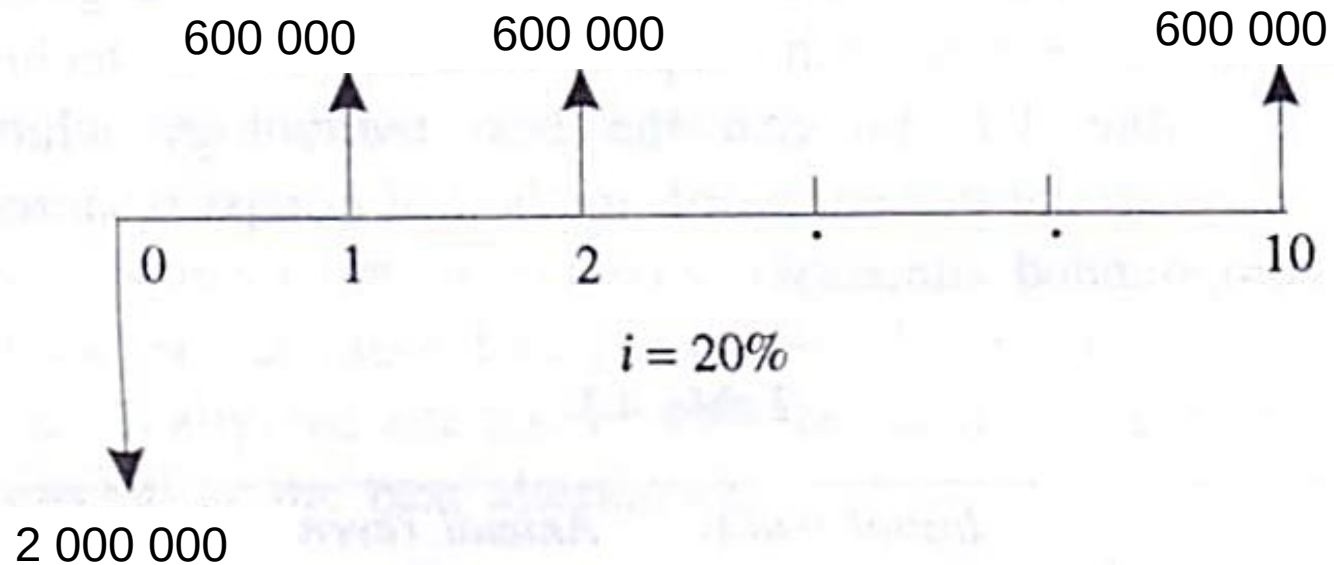
Initial outlay, $P =$ R2 000 000

Annual revenue, $A =$ R600 000

Interest rate, $i = 20\%$, compounded annually

Life of this technology, $n = 10$ years

Cash Flow Diagram: Technology 2



Example 1: Solution Contd...(Technology 2)

$$\begin{aligned}PW(20\%)_2 &= -2\,000\,000 + 600\,000 \times (P/A, 20\%, 10) \\&= -2\,000\,000 + 600\,000 \times (4.1925) \\&= -2\,000\,000 + 2\,515\,500 \\&= \text{R}515\,500\end{aligned}$$

Example 1 (Technology 3)

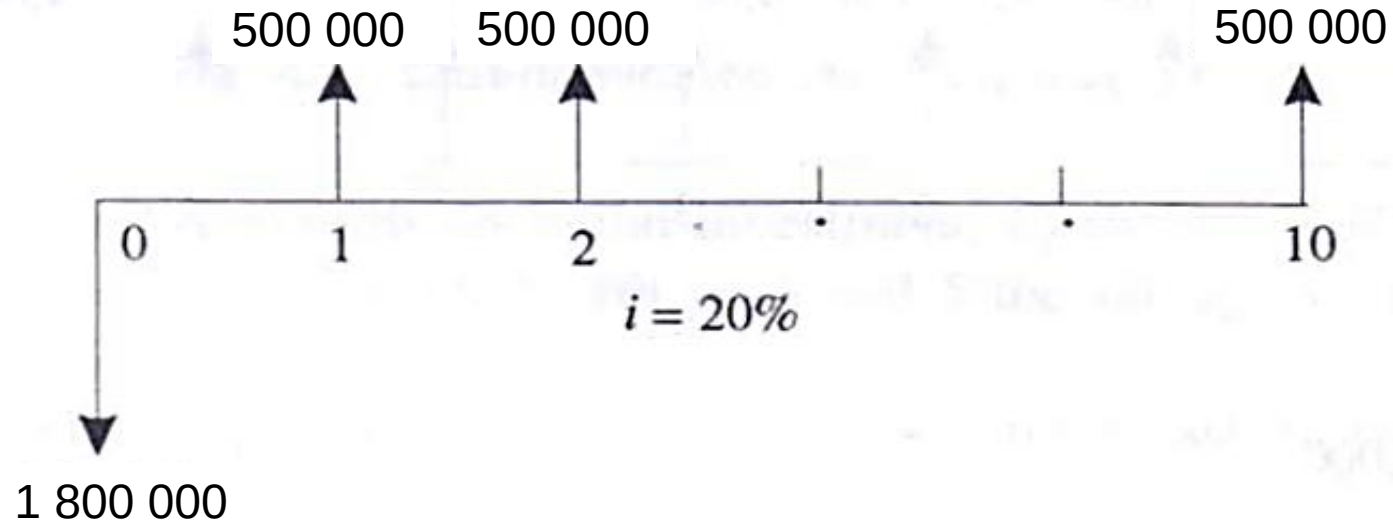
Initial outlay, $P =$ R1 800 000

Annual revenue, $A =$ R500 000

Interest rate, $i = 20\%$, compounded annually

Life of this technology, $n = 10$ years

Cash Flow Diagram: Technology 3



Example 1: Solution Contd...(Technology 3)

$$\begin{aligned}PW(20\%)_3 &= -1\,800\,000 + 500\,000 \times (P/A, 20\%, 10) \\&= -1\,800\,000 + 500\,000 \times (4.1925) \\&= -1\,800\,000 + 2\,096\,250 \\&= R296\,250\end{aligned}$$

Decision on Example 1

Since the NPV of technology 2 is the highest, choose technology 2

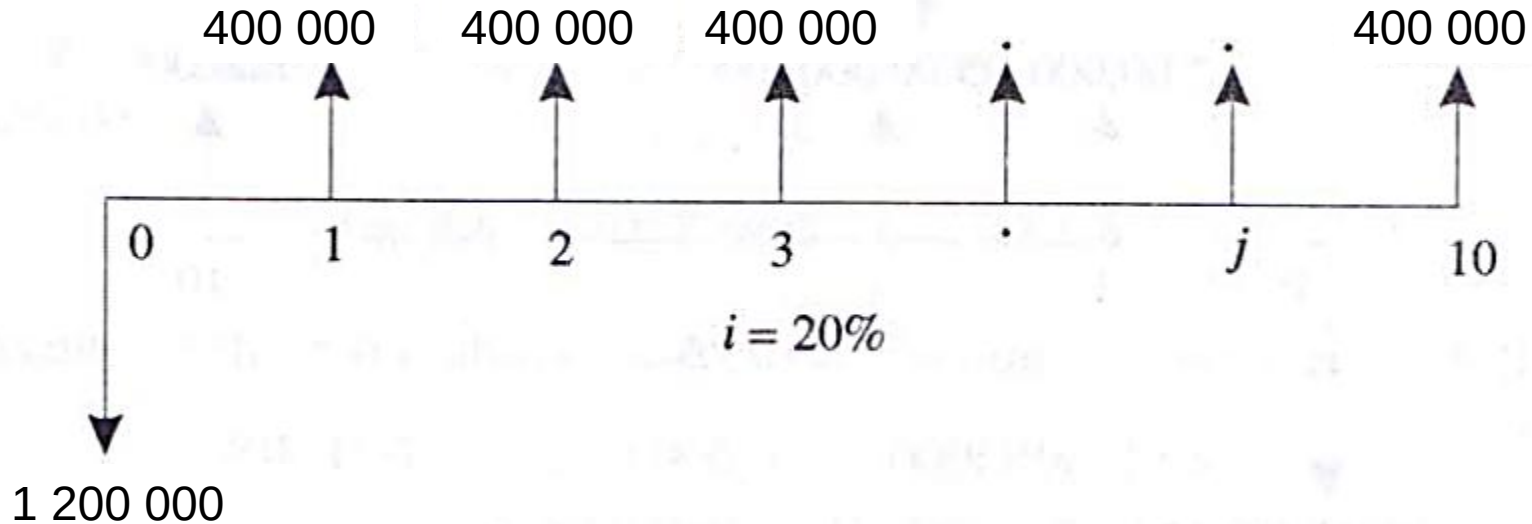
Example 1: Comparing Present and Future worth techniques for the same problem

VW South Africa is planning to expand its production operation. It has identified three different technologies for meeting the goal. The initial outlay and annual revenues with respect to each of the technologies are presented in the table below. Suggest the best technology which is to be implemented based on the present worth method of comparison assuming 20% interest rate, compounded annually.

Example 1 Contd.....

	<i>Initial outlay</i>	<i>Annual revenue</i>	<i>Life</i>
	R	R	(years)
Technology 1	1 200 000	400 000	10
Technology 2	2 000 000	600 000	10
Technology 3	1 800 000	500 000	10

Example 1 Contd....: Cash Flow Diagram (Technology 1)



Example 1: Solution Contd...(Technology 1) **Future Worth**

The present worth analysis for the same problem is given as:

$$\begin{aligned} PW(20\%)_1 = & -1\,200\,000 + 400\,000 \times (P/F, 20\%, 1) + 400\,000 \times (P/F, 20\%, 2) + \\ & 400\,000 \times (P/F, 20\%, 3) + 400\,000 \times (P/F, 20\%, 4) + 400\,000 \times (P/F, 20\%, 5) + \\ & 400\,000 \times (P/F, 20\%, 6) + 400\,000 \times (P/F, 20\%, 7) + 400\,000 \times (P/F, 20\%, 8) + \\ & 400\,000 \times (P/F, 20\%, 9) + 400\,000 \times (P/F, 20\%, 10) \end{aligned}$$

Example 1: Solution Contd...(Technology 1) **Assuming Future Worth Solution**

$$PW(20\%)_1 = -1\,200\,000 + (400\,000 \times 0.8333) + (400\,000 \times 0.6944) + (400\,000 \times 0.5787) + (400\,000 \times 0.4823) + (400\,000 \times 0.4019) + (400\,000 \times 0.3349) + (400\,000 \times 0.2791) + (400\,000 \times 0.2326) + (400\,000 \times 0.1938) + (400\,000 \times 0.1615)$$

$$PW(20\%)_1 = [-1\,200\,000 + 333320 + 277760 + 231480 + 192920 + 160760 + 133960 + 111640 + 93040 + 77520 + 64600] = \mathbf{477000}$$

Decision: For a revenue dominated cash flow, if the discounted value for the analysed cash flow >0 , it implies that project option is viable.

Example 2

An engineer has two bids for an elevator to be installed in a new building. The details of the bids for the elevators are as presented in the following slide:

Example 2: Tabulation

<i>Bid</i>	<i>Engineer's estimates</i>		
	<i>Initial cost</i>	<i>Service life (years)</i>	<i>Annual operations & maintenance cost</i>
	R		R
Alpha Elevator Inc.	R450000	15	R27000
Beta Elevator Inc.	R540000	15	R28500

Determine which bid should be accepted based on the present worth method of comparison assuming 15% interest rate compounded annually

Example 2: Solution 1

Bid 1

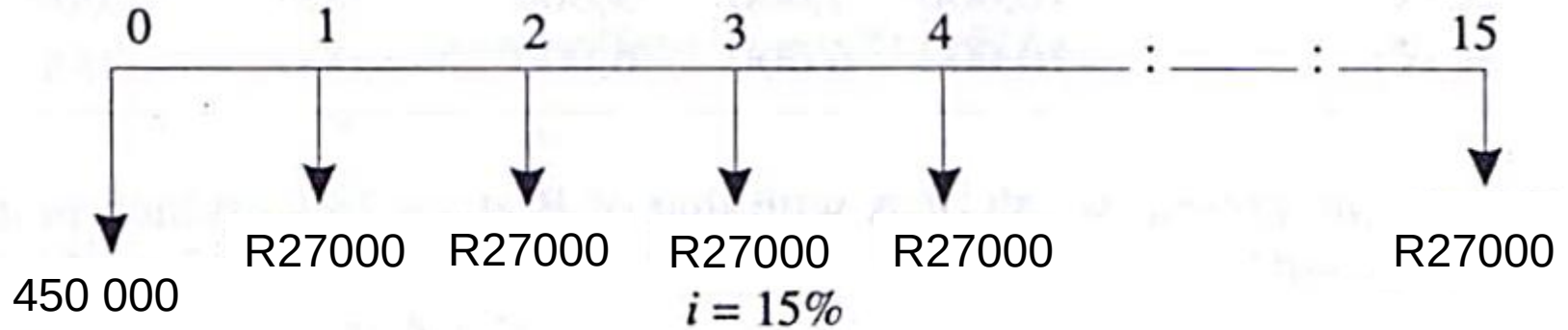
Initial cost, $P =$ R450 000

Annual operation and maintenance cost, $A =$ R27 000

Life = 15 years

Interest rate, $i = 15\%$, compounded annually.

Example 2: Cash Flow Diagram (Solution 1)



Example 2: Solution 1.....Contd.

$$\begin{aligned}PW(15\%) &= 450\,000 + 27,000(P/A, 15\%, 15) \\&= 450\,000 + 27,000 \times 5.8474 \\&= 450\,000 + 157\,879.80 \\&= R607\,879.80\end{aligned}$$

Example 2: Solution 2

Bid 2

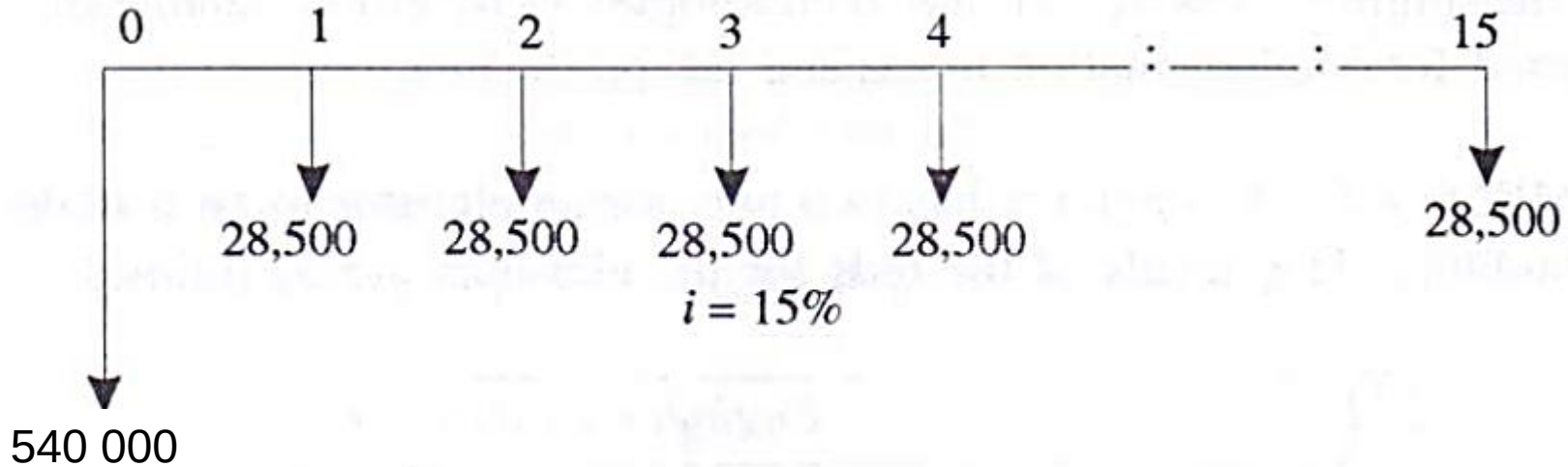
Initial cost, $P =$ R540 000

Annual operation and maintenance cost, $A =$ R28 000

Life = 15 years

Interest rate, $i = 15\%$, compounded annually.

Example 2: Cash Flow Diagram (Solution 2)



Cash flow diagram for bid 2

Example 2: Solution 2.....Contd.

$$PW(15\%) = R540000 + R28500 \times (P/A, 15\%, 15)$$

$$= R540000 + R28500 \times 5.8474$$

$$= R540000 + R166650.90$$

$$= R706\,650.90$$

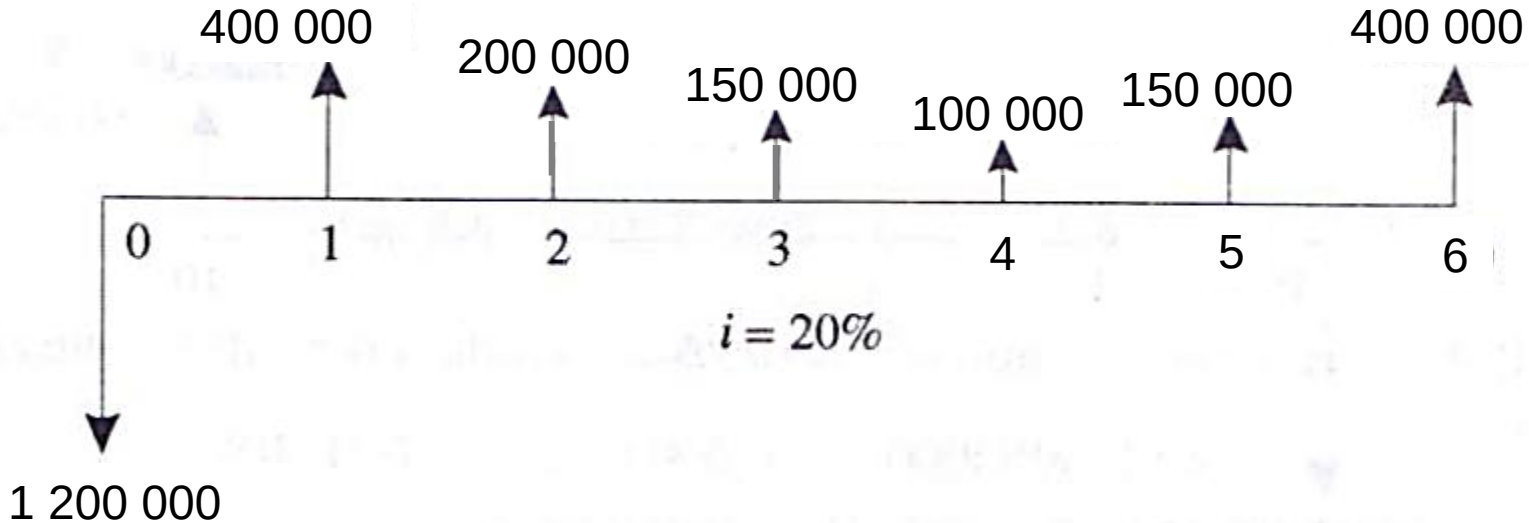
Decision on Example 2

The total present worth cost of bid 1 is less than that of bid 2. Hence, bid 1 is to be selected for implementation. That is the elevator from Alpha Elevator Inc. is to be purchased and installed.

Revenue Dominated Cash Flow with “mixed amount”: (Discounting to the Present from the Future)

Considering the VW South Africa example above, assuming a slight tweak is brought onboard the cash flow scenario such that the generated revenue is not constant but with a set of mixed income as presented in the cash flow diagram in the following slide, you are required to carry out a present worth analysis of the investment at the same interest rate of 20% compounded annually.

Example 2: Cash Flow Diagram (Technology 1 (Amended))



Example 2: Solution Contd...(Technology 1) **Future Worth Solution Only**

The present worth analysis for the same problem is given as:

$$PW(20\%)_1 = -1\,200\,000 + 400\,000 \times (P/F, 20\%, 1) + 200\,000 \times (P/F, 20\%, 2) + 150\,000 \times (P/F, 20\%, 3) + 100\,000 \times (P/F, 20\%, 4) + 150\,000 \times (P/F, 20\%, 5) + 400\,000 \times (P/F, 20\%, 6)$$

Example 2: Solution Contd...(Technology 1) **Future Worth Solution Only**

$$PW(20\%)_1 = -1\,200\,000 + (400\,000 \times 0.8333) + (200\,000 \times 0.6944) + (150\,000 \times 0.5787) + (100\,000 \times 0.4823) + (150\,000 \times 0.4019) + (400\,000 \times 0.3349)$$

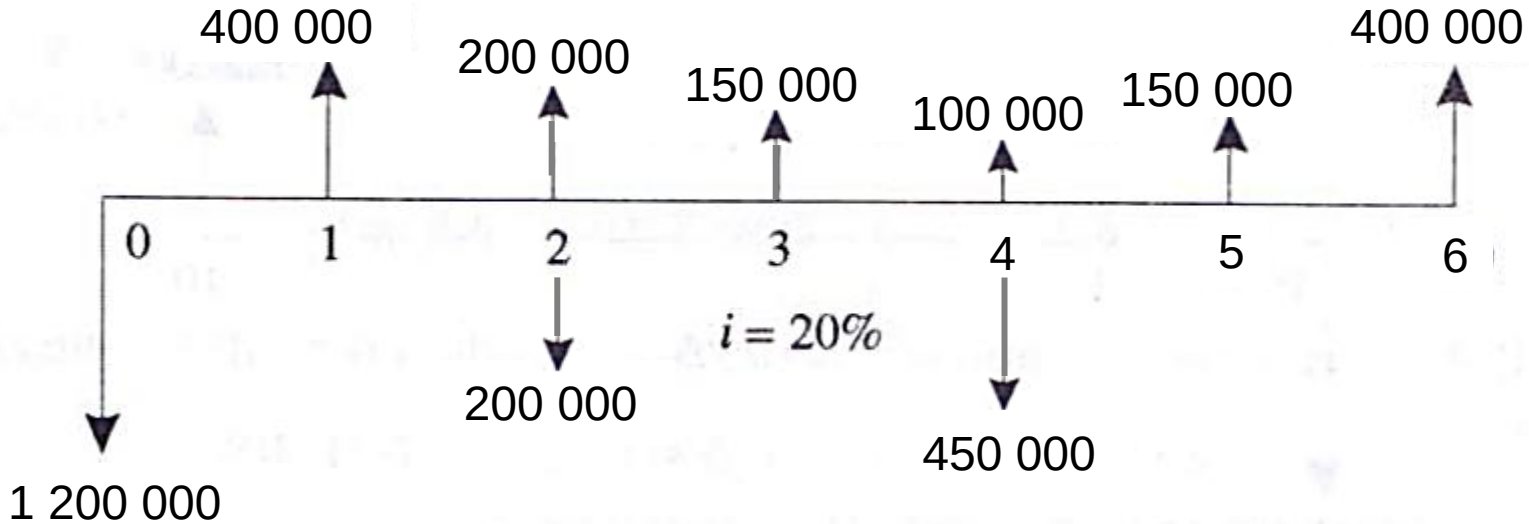
$$PW(20\%)_1 = [-1\,200\,000 + 333\,320 + 138\,880 + 86\,805 + 48\,230 + 60\,285 + 133\,960] = -398\,520$$

Based on the negative output of this project's net cash flow, it is considered not viable.

Revenue Dominated Cash Flow with “mixed amount”: (Discounting to the Present from the Future)

Considering the VW South Africa example above, assuming a slight tweak is brought onboard the cash flow scenario such that the generated revenue is not constant but with a set of mixed income as presented in the cash flow diagram in the following slide below, you are required to carry out a present worth analysis of the investment at the same interest rate of 20% compounded annually.

Example 3: Cash Flow Diagram (Technology 1) **Modified**



Example 3: Solution Contd...(Technology 1) **Future Worth Solution Only**

The present worth analysis for the same problem is given as:

$$PW(20\%)_1 = -1\,200\,000 - 200\,000 \times (P/F, 20\%, 2) - 450\,000 \times (P/F, 20\%, 4) + 400\,000 \times (P/F, 20\%, 1) + 200\,000 \times (P/F, 20\%, 2) + 150\,000 \times (P/F, 20\%, 3) + 100\,000 \times (P/F, 20\%, 4) + 150\,000 \times (P/F, 20\%, 5) + 400\,000 \times (P/F, 20\%, 6)$$

Example 3: Solution Contd...(Technology 1) **Future Worth Solution Only**

$$PW(20\%)_1 = -1\,200\,000 - (200\,000 \times 0.6944) - (450\,000 \times 0.4823) + (400\,000 \times 0.8333) + (200\,000 \times 0.6944) + (150\,000 \times 0.5787) + (100\,000 \times 0.4823) + (150\,000 \times 0.4019) + (400\,000 \times 0.3349)$$

$$PW(20\%)_1 = [-1\,200\,000 - 138880 - 217035 + 333320 + 138880 + 86805 + 48230 + 60285 + 133960] = \mathbf{-754435}$$

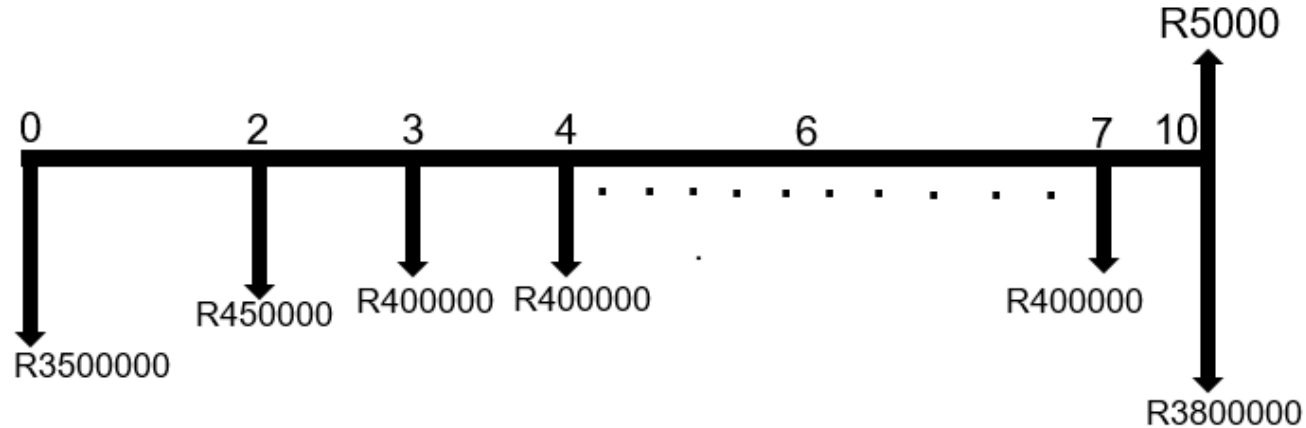
Based on the negative output of this project's cash flow, it is considered not viable.

Example 4: Mixed Scenario Problem-[Present Worth Analysis Determination]

As a client making some major purchases for an advanced engineering laboratory from a company, you are required to make an initial down payment of R3500000, pay R450000 at the end of the second year, make a payment of R400000 from the end of the third year to the end of the seventh year and make a closing payment of R3800000 at the end of the tenth year.

Assuming client loyalty to consistent payments will result in a gift of R5000 at the end of the tenth year, you are required to determine the present worth analysis of the entire cash flow. The interest rate is taken to be 12% compounded annually.

Example 4: Mixed Scenario Problem-[Present Worth Analysis Determination]



Example 4: Mixed Scenario Problem-[Present Worth Analysis Determination]

$$-\left(3500000 + (450000(P/F, 12\%, 2)) + (400000(P/A, 12\%, 5))\right) \\ * (P/F, 12\%, 2) + 3800000(P/F, 12\%, 10) + 5000(P/F, 12\%, 10)$$

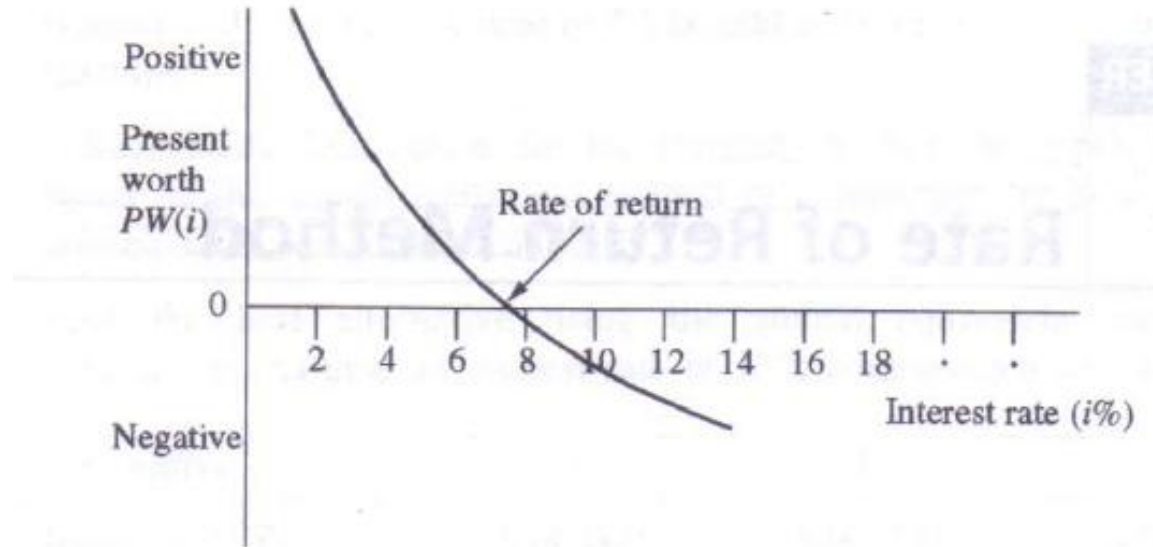
$$=-(3500000+(450000*0,7972)+(400000*3,6048*0,7972)+(3800000*0,322))+(5000*0,322)$$

$$= -6230228,6240$$

Exercise #3: Mixed Scenario Problem-[Present Worth Analysis Determination]

The Incredible Connection store at Menlyn, South Africa recently advertised its latest Sony 77-inch 4K OLED TV that comes with a free carpet cleaner all for R80000. It is such that the TV brand can be taken home by the customer without making any initial down payment however, a first payment of R65000 is required at the end of the second year **while an equal yearly payment of R8000 is expected to be made from the end of the third year for a period of 10 years.** The interest rate in this scenario is given as 12% which is compounded annually. You are required to sketch the cash flow diagram and determine the present worth analysis for this deal.

Rate of Return Graph



Example 4: Rate of Return

A person is planning a new business. The cash flow for the new business is as presented below. The expected life of the business is five years. Find the rate of return for the business.

Period	0	1	2	3	4	5
Cash Flow	-R100000	R30000	R30000	R30000	R30000	R30000

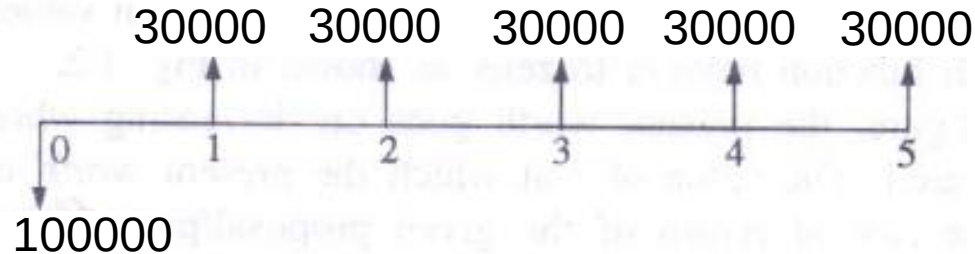
Solution

Initial Investment=R100000

Annual equal revenue=R30000

Duration=5 years

Cash Flow Diagram



Solution

The present worth function for the business is given as:

$$PV(i) = -100000 + 30000(P/A, i, 5)$$

When $i = 10\%$:

$$PV(10\%) = -100000 + 30000(P/A, 10\%, 5)$$

$$= -100000 + 30000(3.7908)$$

$$= R13724$$

Solution

When $i = 15\%$:

$$\begin{aligned} PV(15\%) &= -100000 + 30000(P/A, 15\%, 5) \\ &= -100000 + 30000(3.3522) \\ &= R566 \end{aligned}$$

When $i = 18\%$:

$$\begin{aligned} PV(18\%) &= -100000 + 30000(P/A, 18\%, 5) \\ &= -100000 + 30000(3.1272) \\ &= R-6184 \end{aligned}$$

Solution

By interpolating:

$$15\% = 566$$

$$i\% = 0$$

$$18\% = -6184$$

$$\frac{i - 15\%}{0 - 566} = \frac{18 - i}{-6184 - 0}$$

Solution

$$i = 15\% + \frac{566 - 0}{566 - (-6184)} * (3\%)$$

$$= 15\% + 0.252\%$$

$$= 15.252\%$$

Exercise 4: Rate of Return

A company is trying to diversify its business into a new product line. The life of the project is 10 years with no salvage value at the end of its life. The table below presents the cash distribution over the years. You are required to find the rate of return.

[illegible]

Applying the knowledge of time value of money to the semester project including the **MARR and project viability**

Learning Outcomes Evaluated

- Can you now define and introduce the rudiments of cash flow diagrams
- Can you discuss, analyse, apply, and evaluate the time value of money (with focus on discounting to the present, to obtain the Present Value (PV) of money
- Can you establish practical relationship between the time value of money, the MARR and a Project's viability

The End

Next Topic

Unit 13: Systems thinking & business structure integratedness



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